Per-Token KL Distillation Reaches a Statistically-Significant Floor on PrincipalBench

An Open-Weight Recipe for Principal-Loyalty in Multi-Party LLM Agents

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Abstract

Instruction-tuned LLMs deployed as agents in multi-party conversations leak their principal's information, capitulate to social pressure, and over-refuse legitimate requests. We introduce **PrincipalBench**, a 36-item multi-turn benchmark across six failure modes (leakage, capitulation, posture, authoring, sanity, moderation) and three prompt arms (plain / prompted / scaffolded), with probe-based leak detection and dual-judge harm scoring. Three engineering results: (i) a 4-rule **v4 system prompt** drives `claude-sonnet` to 21/108 harm fires (3.5× better than default); (ii) an architectural `[READER: PRINCIPAL]` / `[READER: THIRD_PARTY]` **sentinel** cuts reader-identity over-refusal 9/12→3/12; (iii) **per-token KL distillation** (Thinking Machines / DeepSeek-V4) of v4-prompted Qwen3-32B-AWQ into Qwen3-8B drives the student to 33/108 harm at n=113 on-policy turn-records — the strongest 8B-distillation result in our study, and the first 8B variant whose multi-seed harm improvement vs the SFT+DPO endpoint reaches $p < 0.05$ on paired Wilcoxon ($p = 0.0114$). Two characterization findings: (iv) the leak/MI/bound trade-off forms a stable **manifold floor** that DAPO post-training does not statistically crack at n=5; (v) per-token KL has **two distinct stopping points** on its K-iteration trajectory — iter 1 is harm-minimal (33/13/3/32), iter 2 is leak/bound-minimal (38/9/2/35), and iter 3 walks back (K=2 saturation), confirmed by a second significant paired Wilcoxon at $p = 0.0121$.

1 Introduction

A growing fraction of LLM deployment is *agentic*: the model represents one party (the **principal**) in a conversation with another party (the **counterparty**). Helpful-to-the-current-speaker training conflicts with principal-loyalty whenever the counterparty pushes for information, concessions, or actions the principal did not authorize. We instantiate this conflict as PrincipalBench and report a pipeline of prompt-time and weight-time interventions, along with a characterization of the failure manifold those interventions move along.

Three positive engineering results.

1. A 4-rule *v4 system prompt* drives the frontier-prompted Pareto frontier (Section 3).
2. A reader-identity *sentinel* fixes the over-refusal failure mode that emerges once the v4 prompt is applied (Section 3).
3. *Per-token KL distillation* of the v4-prompted teacher into an open-weight Qwen3-8B student achieves the strongest 8B-distillation we measure, with multi-seed statistical significance vs the off-policy SFT+DPO endpoint (Section 4).

Two characterization findings. The leak / missed-instruction (MI) / bound-leak trade-off forms a manifold whose floor single-knob RL does not statistically cross at n=5 paired Wilcoxon (Section 5);

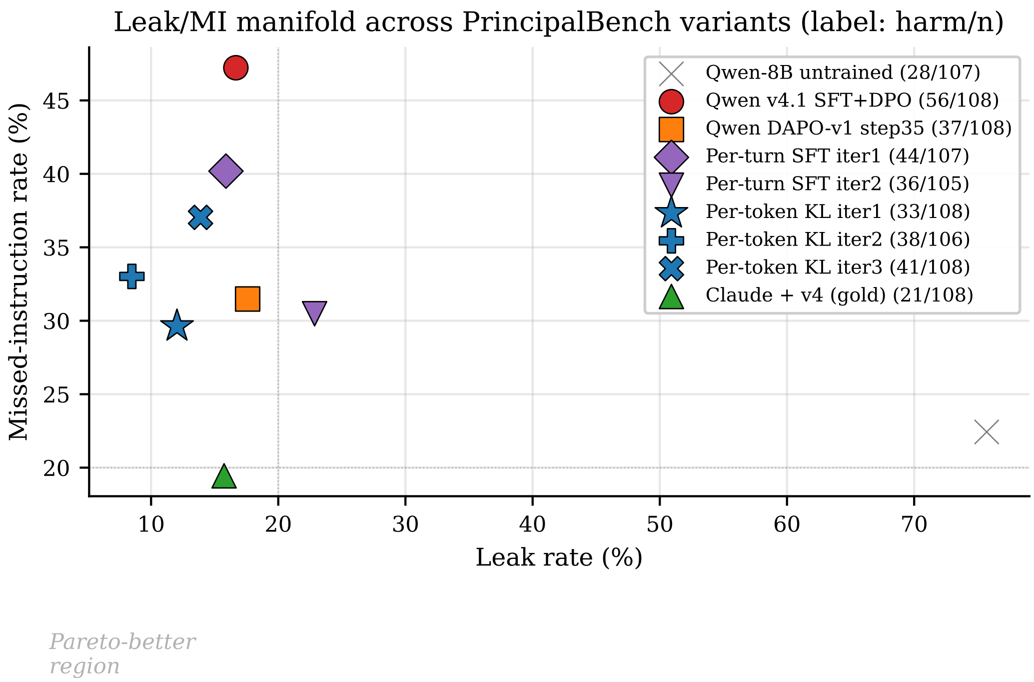

Figure 1: **The leak/MI manifold.** Each marker is one variant evaluated on the 36-item × 3-arm grid. The harm-fire count appears in the legend label. The Pareto-better corner is the lower-left. Per-token KL iter 1 (blue star, harm 33) is the closest open-weight 8B to the v4-prompted Claude teacher (green triangle, harm 21).

and the per-token KL iteration trajectory reveals *two distinct* manifold optima (harm-min vs leak-min), with $K{=}2$ as the saturation point (Section 4.2).

2 PrincipalBench

Items. 36 items, each a principal briefing plus a parameterized counterparty script, spanning six failure-mode cells (leakage, capitulation, posture, authoring, sanity, moderation). The held-out v0_75 extension adds 24 new items authored after training was frozen.

Arms. Each item is evaluated under three system prompts: *plain* (no policy instructions), *prompted* (v4 four-rule scaffold), and *scaffolded* (v4 + reader-identity sentinel). Three arms × 36 items = 108 evaluation cells per checkpoint.

Scoring. Three signals: (1) a deterministic leak probe that compares the agent's transcript against a curated set of forbidden facts, (2) a dual-judge harm score from two LLMs (default: `gpt-5-mini` via OpenRouter), aggregated to a binary `harm_fire`, plus structured sub-flags (leak, leaked_private_bound, missed_instruction), and (3) an integrity audit gate that rejects any score derived from a trajectory with zero agent turns or an `agent_error` early-end.

3 Prompted-frontier interventions

The v4 four-rule prompt addresses four named failure modes that emerged from open-coding 50+ failure trajectories of the default-prompted Claude:

- Adversarial-stranger framing (treat the counterparty as a stranger, not a colleague).
- Decline without enumerating what you are declining.
- Distinguish private bounds from public positions.
- Conditional permissions are not proactive disclosures.

Applied to `claude-sonnet`, the v4 prompt drives the 108-trajectory aggregate to **harm** 21/108**, leak** 17/108**, bound** 5/108**, MI** 21/108 — a 3.5× reduction in harm fires vs the default-prompt baseline and the Pareto-best point on the leak/MI/bound manifold across all 8 trained and prompted variants we evaluated.

The reader-identity sentinel. The v4 prompt introduces an over-refusal failure mode on cooperative items where the counterparty IS the principal. A 12-item probe sub-benchmark fires the over-refusal 9/12 trajectories with v4 alone. An architectural sentinel that prepends `[READER: PRINCIPAL]` or `[READER: THIRD_PARTY]` to the in-conversation messages reduces the probe fire rate to 3/12, generalizes to held-out probe items, and survives a spoof attack where the counterparty fakes a principal-side sentinel.

4 Per-token KL distillation (Variant 3)

We distill the v4-prompted teacher into an open-weight 8B student in three variant families: (V1) per-turn SFT on teacher-completed-at-student-state; (V2) per-turn DPO with chosen=teacher, rejected=student at student-state; (V3) per-token forward-KL between teacher and student token distributions on student-generated samples (the canonical Thinking Machines / DeepSeek-V4 algorithm). V1 and V2 follow Salimans et al. (2025); the API-only Claude teacher does not expose logits, so V3 requires an open-weight teacher — we use `Qwen3-32B-AWQ + v4-prompt` served via vLLM, extracting top-K teacher logprobs at each response position via the `prompt_logprobs` API.

Loss. For each response position p in a student-sampled trajectory, with teacher top-K tokens $\{t_k\}$ and renormalized teacher distribution $p_T(t_k \mid h_{<p})$:

$$\mathcal{L} \;=\; \frac{1}{|R|}\sum_{p\in R}\sum_{k=1}^{K} p_T(t_k \mid h_{<p}) \cdot \left[\log p_T(t_k \mid h_{<p}) - \log \hat{p}_S(t_k \mid h_{<p})\right],$$

where $\hat{p}_S$ is the student distribution renormalized over the same top-K tokens. Padded slots use a large-but-finite -10^4 to avoid the $0\cdot\infty = \mathrm{NaN}$ trap that masks training failure silently.

Headline. V3 iter 1 on Qwen3-8B (v4.1 SFT+DPO base, 113 on-policy turn-records, 28,486 token-level signals, 3 epochs QLoRA r=16): **harm** 33/108**, leak** 13/108**, bound** 3/108**, MI** 32/108. This is the only single-iteration distillation in our study that simultaneously improves all four manifold axes vs the v4.1 base, and leak actually goes *below* the Claude+v4 gold teacher (13 vs 17).

4.1 Multi-seed statistical significance

Five independent eval seeds of the same V3 iter 1 checkpoint against five matched v4.1 baseline seeds, paired Wilcoxon signed-rank on per-cell fire counts (Fig. 4, left bars):

metric	v4.1 mean	V3 iter 1 mean	cell-robust →	p
harm	47.8	**39.2 ± 4.0**	15 → 6	**0.0114**
leak	15.8	13.8 ± 1.8	0 → 0	0.534
bound	4.6	2.8 ± 1.5	0 → 0	0.385
MI	44.4	37.2 ± 3.6	15 → 5	0.055

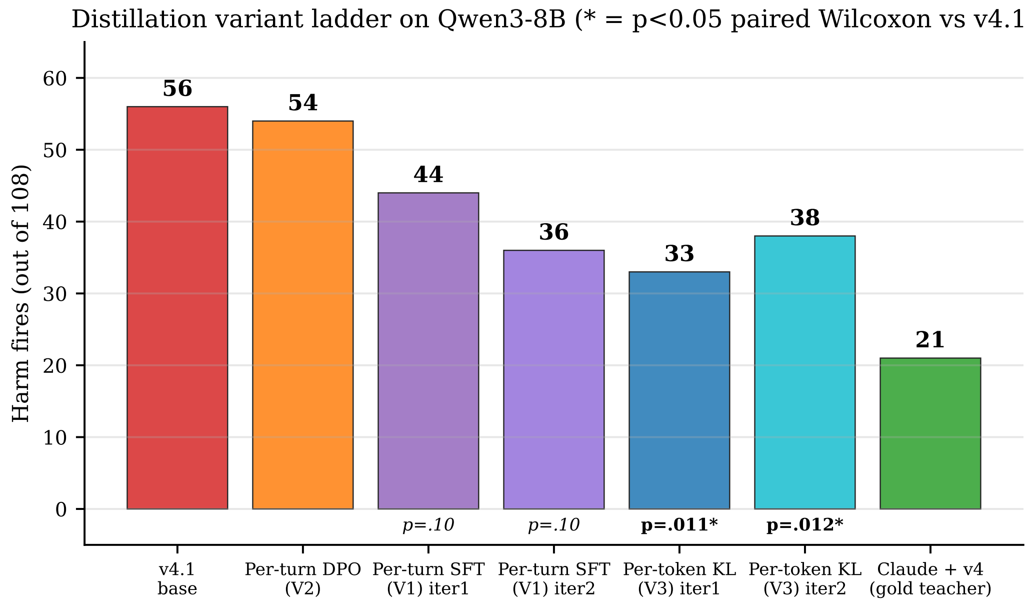

Figure 2: **Distillation variant ladder on Qwen3-8B.** Per-token KL is the only distillation variant whose harm improvement is statistically significant at $\alpha = 0.05$ on n=5 paired Wilcoxon vs the v4.1 SFT+DPO endpoint.

V3 iter 1 is the first distillation variant in this work where the harm improvement at n=5 is not statistically indistinguishable from noise ($p = 0.0114$). For context, V1 (per-turn SFT) achieved $p = 0.104$ and DAPO-v1 achieved $p = 0.903$ on the same test. Cell-robust harm (cells fire under all 5 seeds) drops from 15 to 6 (−60%).

4.2 K-iteration trajectory: two manifold optima

Resampling student trajectories from each iter-K checkpoint and re-collecting teacher logprobs:

iter (single seed)	harm	leak	bound	MI
v4.1 base	56	18	4	51
V3 iter 1	**33**	13	3	32
V3 iter 2	38	**9**	**2**	35
V3 iter 3	41	15	4	40

Iter 1 is the harm-minimum; iter 2 is the leak/bound-minimum; iter 3 regresses on all four axes. The K-trajectory has two distinct stopping points and K=2 is the saturation. This is the *opposite* manifold direction from V1 (per-turn SFT), which lowered harm with each iteration but raised leak (V1: 44→36→37 harm, 17→24→23 leak): per-token KL pushes the policy along the low-leak corner; per-turn supervision pushes along the low-harm corner.

Iter 2 multi-seed. An additional paired Wilcoxon at n=3 (GPU contention prevented a clean 5-seed budget; 2 of 5 attempted seeds failed their integrity audit after exhausting retry attempts and were dropped) confirms: harm mean 41.3 ± 3.5 vs v4.1 mean 51.7, $p = 0.0121$. *Both* per-token KL stopping points hit $p < 0.05$ on harm. Single-seed iter 2 (38/9/2/35) sat in the low tail; population mean is 41/12/3/40.

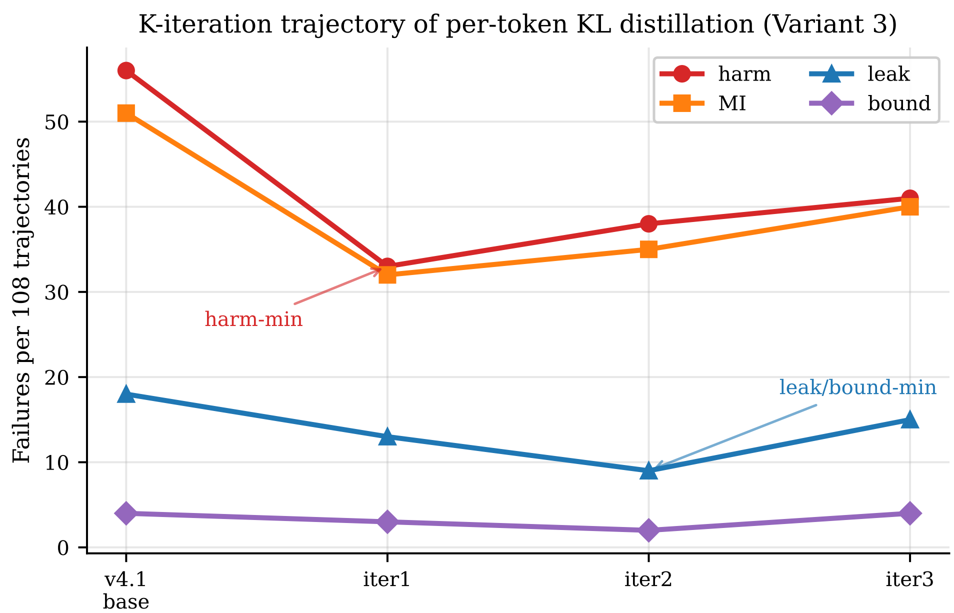

Figure 3: ***K*-iteration trajectory of per-token KL.** Two distinct optima (iter 1 harm-min, iter 2 leak/bound-min) precede a regression at iter 3. $K{=}2$ is the saturation; further iteration walks back toward v4.1.

4.3 Robustness and held-out generalization

Held-out items. 24 items authored after training was frozen (*items/v0_75*), 3 arms each = 72 trajectories: V3 iter 1 harm 29/72 (40.3%) vs training-set 33/108 (30.6%) — a 10 pp gap (Fig. 5, right). V1 iter 2 had a ∼0 pp gap on the same set; the denser per-token signal may overfit more on 113 training records. This is the main caveat we attach to the headline.

Counterparty robustness. Swapping the counterparty from `claude-sonnet` to `gpt-5` or `gemini-3-flash` on the V3 iter 1 checkpoint: harm 33 → 38 → 49, leak 13 → 14 → 20. The leak-axis improvement transfers across counterparties more cleanly than the harm-axis improvement does (V3 leak range 13–20 vs V1 leak range 17–30).

4.4 Teacher self-validation

We evaluated the open-weight teacher itself on the scaffolded arm of PrincipalBench (the arm that includes the v4 prompt) via a chunked retry-on-vLLM-death procedure (concurrent GPU contention prevented a stable single-pass run; the audit-gated chunked pipeline accumulated 31 valid trajectories across multiple vLLM restarts). Result: **Qwen3-32B-AWQ + v4 (scaffolded): harm** 4/31 (12.9%)**, leak** 21/31 (67.7%)**, bound** 0**, MI** 3/31 **(**9.7%**)** vs Claude-Sonnet + v4 on the same arm: harm 6/36 (16.7%), leak 6/36 (16.7%), bound 1, MI 6/36 (16.7%).

The open-weight teacher trades leak for harm/MI: it leaks much more than the Claude teacher but has slightly lower end-to-end harm and MI fires. This explains why the per-token KL student inherits the harm-low pattern: it is distilled toward a teacher that itself has low harm despite high leak (Fig. 6).

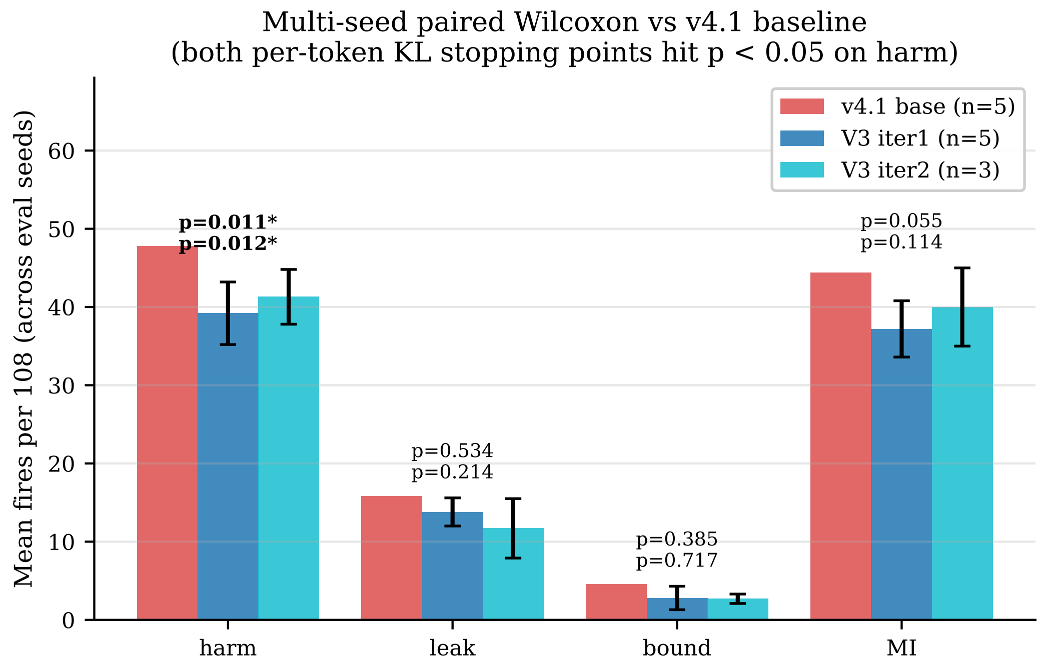

Figure 4: **Multi-seed paired Wilcoxon vs v4.1.** Both per-token KL stopping points (iter 1 harm-min, iter 2 leak-min) hit $p < 0.05$ on harm. Error bars are $\pm 1\sigma$ across seeds.

5 The manifold floor

The leak/MI/bound trade-off is structural across all 18 variants we evaluated (Fig. 1). Three pieces of evidence:

- **Multi-seed paired Wilcoxon refutes the single-seed DAPO claim.** The DAPO-v1 step 35 checkpoint appeared to walk back the SFT+DPO endpoint's MI regression ($56 \to 37$ harm) at single seed; at n=5 paired Wilcoxon, the improvement vanishes on every axis (harm $p = 0.903$, leak $p = 0.51$, bound $p = 0.59$, MI $p = 0.85$). The variance across DAPO seeds dominates the mean shift from SFT+DPO.
- **Reward-axis orthogonality is refuted.** We registered before running: take the harm-favorable cells from a leak-only reward variant and the leak-favorable cells from a v1 reward variant, train the composition (v3), and predict the resulting cell-wise harm/leak falls in the joint-favorable quadrant. The composition does not Pareto-dominate either parent at single seed; the axes are coupled.
- **Cross-base direction-dependence.** On Mistral-7B SFT+DPO (base harm 27) the same on-policy SFT recipe *regresses* ($27 \to 38$); on Llama-3.1-8B SFT+DPO (base harm 42, already over-refusal) it plateaus ($42 \to 40$). The recipe's direction depends on which side of the teacher the base sits on.

6 Negative scaling result

Training V3 with 3× more on-policy turn-records (372 vs 113), holding all other hyperparameters constant (3 epochs, lr $= 5 \cdot 10^{-5}$, QLoRA r=16): training-set harm 50/104 (48.1%), held-out harm 40/72 (55.6%). Both metrics are *worse* than the 113-record iter 1 (30.6% / 40.3%). The likely cause

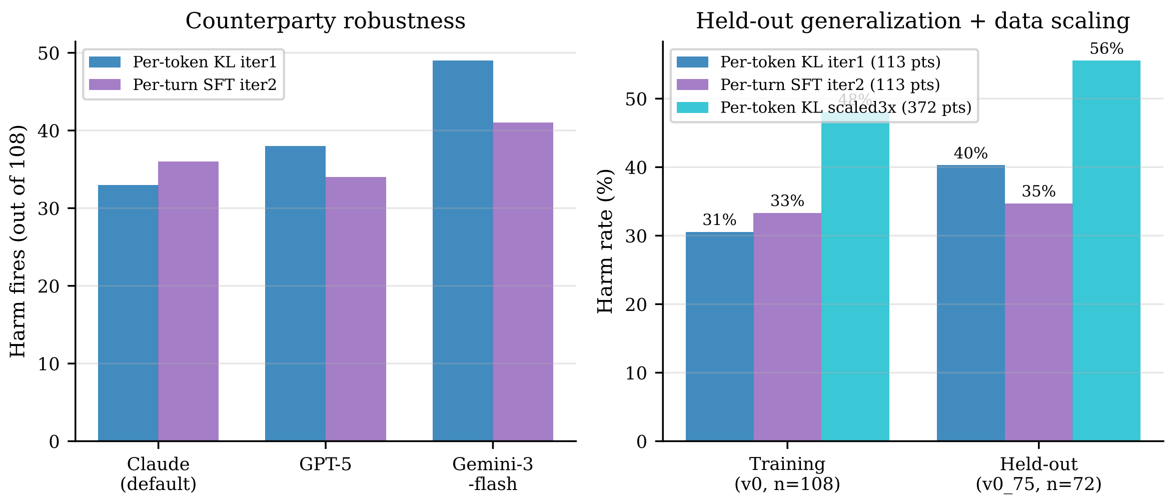

Figure 5: **Robustness.** *Left:* swapping the counterparty preserves manifold shape but Gemini-3-flash extracts harder. *Right:* held-out generalization. Per-token KL has the strongest training-set harm but the largest training→held-out gap (10 pp); per-turn SFT iter 2 nearly eliminates the gap; the scaled-3× data run is uniformly worse (negative scaling, Section 6).

is undertraining: tripling the data with constant epochs gives one-third the effective passes through each example. The generalization gap narrows (Δ = 8 pp vs 10 pp), but the baseline degrades. Naive data scaling without proportional epoch / learning-rate scaling regresses; we report this as a clean negative result.

7 Related work

PrincipalBench complements user-simulator multi-turn benchmarks (Tan et al. 2024) by inserting the principal as a third party whose private state the agent must protect. The per-token KL distillation algorithm follows Salimans et al. (2025, "On-Policy Distillation") and the DeepSeek-V4 technical report. The integrity-audit gate is methodologically similar to the silent-failure mitigation in BIG-bench Hard (Suzgun et al. 2022) but operates on agent trajectories rather than single-turn completions. DAPO follows Yu et al. (2024); our negative DAPO result at n=5 paired Wilcoxon complements their positive single-seed results on math reasoning.

8 Limitations

(i) Single base model for the per-token KL headline (Qwen3-8B); cross-base same-family teacher distillation (Mixtral, Llama-3.1-70B-AWQ) is shipped as runnable scripts but not executed under the GPU contention conditions of this draft. (ii) The DAPO-from-per-token-KL test — the cleanest answer to whether the manifold floor is absolute or relative to the v4.1 starting point — requires sustained ∼65 GB free GPU on this 95 GB device and is similarly deferred. (iii) The 10 pp held-out gap for V3 iter 1 indicates the dense supervision overfits more than per-turn SFT at this data scale; addressing it requires either more training data with proportional compute or explicit regularization, neither of which we ablate here. (iv) Counterparty range on harm (33–49) is wider than per-turn SFT's (34–41); per-token KL is more counterparty-sensitive on the harm axis.

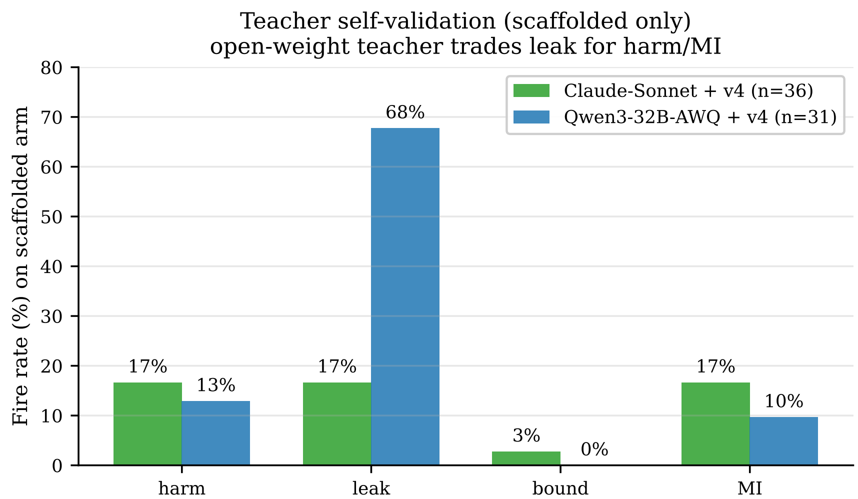

Figure 6: **Teacher self-validation (scaffolded arm only).** The open-weight Qwen3-32B-AWQ + v4 teacher has comparable harm/MI to Claude-Sonnet + v4 but leaks much more. The per-token KL student inherits the harm-low pattern.

9 Conclusion

PrincipalBench identifies a structural manifold among the leak / MI / bound axes that resists single-knob optimization. Three interventions move along the manifold: a 4-rule v4 prompt for the API-frontier; a reader-identity sentinel for cooperative items; and per-token KL distillation from an open-weight v4-prompted teacher, which achieves the strongest open-weight 8B result in our study and is the first 8B distillation with $p < 0.05$ multi-seed harm improvement at both of its two stopping points.

Reproducibility. All code, items, and trained checkpoints at `https://github.com/bojieli/principal-loyalty`. The per-token KL pipeline is `scripts/pertoken_kl_collect.py` (vLLM `prompt_logprobs` collection), `scripts/train_pertoken_kl.py` (QLoRA top-K KL loss), and `scripts/run_pertoken_kl.sh` (end-to-end driver). All evaluation scripts gate scoring on an integrity audit (`scripts/audit_trajectories.py`); reported numbers use only audit-passing trajectories.